

# Removing Nontrivial Type from Rozendaal's Endpoint Lattice Case

Run `fa_banach_001`, agent lane 17

2026-06-20

## Status

This packet gives a candidate full answer, likely valid, to the question after Theorem 6.3 in Rozendaal's paper [1]. The source asks whether the assumption that  $X$  has nontrivial type is necessary in the  $R$ -bounded operator-valued endpoint calculus. It is not necessary: the theorem extends to the endpoint  $p = 1$  for complemented subspaces of  $q$ -concave Banach lattices.

## Source Question

After proving Theorem 6.3, Rozendaal writes:

“We do not know whether the assumption in Theorem 6.3 that  $X$  has nontrivial type is necessary.”

The proof there uses nontrivial type only to obtain, uniformly in  $n$ , a projection from  $L^2([0, 1]; X)$  onto  $\text{Rad}^n(X)$ . We replace that step by a direct lattice square-function model.

## Idea of the Proof

Let  $Y$  be a  $q$ -concave Banach lattice,  $2 \leq q < \infty$ . The finite Rademacher space  $\text{Rad}^n(Y)$  is uniformly isomorphic to the lattice

$$Y(\ell_2^n) = \left\{ (y_1, \dots, y_n) : \|(y_k)\|_{Y(\ell_2^n)} = \left\| \left( \sum_{k=1}^n |y_k|^2 \right)^{1/2} \right\|_Y < \infty \right\}.$$

This is the standard Khintchine square-function equivalence for Banach lattices of finite cotype. Moreover  $Y(\ell_2^n)$  is again  $q$ -concave with constants independent of  $n$ , because pointwise  $\ell_q(\ell_2) \leq \ell_2(\ell_q)$  for  $q \geq 2$ . Thus if  $X$  is complemented in  $Y$ , then  $\text{Rad}^n(X)$  is uniformly complemented in a 1-convex,  $q$ -concave Banach lattice. This is exactly the uniform geometric input required in Rozendaal's proof of Theorem 6.3.

## The Uniform Rademacher Lattice Model

**Lemma 1.** *Let  $Y$  be a  $q$ -concave Banach lattice for some  $2 \leq q < \infty$ . Then for every  $n \in \mathbb{N}$ ,  $\text{Rad}^n(Y)$  is isomorphic, with constants independent of  $n$ , to a 1-convex and  $q$ -concave Banach lattice. More precisely,  $\text{Rad}^n(Y)$  is uniformly isomorphic to  $Y(\ell_2^n)$ , and  $Y(\ell_2^n)$  has  $q$ -concavity constant at most the  $q$ -concavity constant of  $Y$ .*

*Proof.* Since  $Y$  has finite cotype, the standard Khintchine inequality for Banach lattices gives constants independent of  $n$  such that

$$\left( \mathbb{E} \left\| \sum_{k=1}^n r_k y_k \right\|_Y^2 \right)^{1/2} \simeq \left\| \left( \sum_{k=1}^n |y_k|^2 \right)^{1/2} \right\|_Y$$

for all  $y_1, \dots, y_n \in Y$ ; see, for instance, the lattice cotype material in [2]. Hence  $\text{Rad}^n(Y)$  is uniformly isomorphic to  $Y(\ell_2^n)$ .

It remains to record that the lattice constants do not depend on  $n$ . Every Banach lattice is 1-convex. Let  $v^j = (y_{j,k})_{k=1}^n \in Y(\ell_2^n)$ ,  $j = 1, \dots, m$ , and set

$$s_j = \left( \sum_{k=1}^n |y_{j,k}|^2 \right)^{1/2} \in Y_+.$$

If  $C_q(Y)$  is the  $q$ -concavity constant of  $Y$ , then

$$\left( \sum_{j=1}^m \|v^j\|_{Y(\ell_2^n)}^q \right)^{1/q} = \left( \sum_{j=1}^m \|s_j\|_Y^q \right)^{1/q} \leq C_q(Y) \left\| \left( \sum_{j=1}^m s_j^q \right)^{1/q} \right\|_Y.$$

Pointwise, for nonnegative scalars  $a_{j,k}$ ,

$$\left( \sum_j \left( \sum_k a_{j,k}^2 \right)^{q/2} \right)^{1/q} \leq \left( \sum_k \left( \sum_j a_{j,k}^q \right)^{2/q} \right)^{1/2},$$

which is Minkowski's inequality in the form  $\ell_q(\ell_2) \hookrightarrow \ell_2(\ell_q)$  for  $q \geq 2$ . Applying this inside the lattice  $Y$  gives

$$\left\| \left( \sum_j s_j^q \right)^{1/q} \right\|_Y \leq \left\| \left( \sum_{k=1}^n \left( \sum_{j=1}^m |y_{j,k}|^q \right)^{2/q} \right)^{1/2} \right\|_Y = \left\| \left( \sum_{j=1}^m |v^j|^q \right)^{1/q} \right\|_{Y(\ell_2^n)}.$$

Thus  $Y(\ell_2^n)$  is  $q$ -concave with constant at most  $C_q(Y)$ , uniformly in  $n$ .  $\square$

**Lemma 2.** *Let  $X$  be isomorphic to a complemented subspace of a  $q$ -concave Banach lattice  $Y$ , where  $2 \leq q < \infty$ . Then  $\text{Rad}^n(X)$ , uniformly in  $n$ , is isomorphic to a complemented subspace of a 1-convex,  $q$ -concave Banach lattice.*

*Proof.* Identify  $X$  with a complemented subspace of  $Y$ , and let  $P : Y \rightarrow X$  be a bounded projection. The coordinatewise map

$$\tilde{P} \left( \sum_{k=1}^n r_k y_k \right) = \sum_{k=1}^n r_k P y_k$$

is a projection from  $\text{Rad}^n(Y)$  onto  $\text{Rad}^n(X)$ , with  $\|\tilde{P}\| \leq \|P\|$ . Now apply Lemma 1.  $\square$

## Endpoint $R$ -Bounded Calculus

**Theorem 1.** *Let  $X$  be isomorphic to a complemented subspace of a  $q$ -concave Banach lattice for some  $2 \leq q < \infty$ . Let  $-iA$  generate a  $C_0$ -group  $(U(s))_{s \in \mathbb{R}} \subseteq \mathcal{L}(X)$ , let  $\lambda > \omega > \theta(U)$ , and let  $\mathcal{A} \subseteq \mathcal{L}(X)$  be the algebra of bounded operators commuting with  $A$ . Then there exists a constant  $C \geq 0$  such that*

$$R_{D((\lambda+iA)^{1-1/q}),X}(\{f(A) : f \in R\mathcal{H}^\infty(\text{St}_\omega; \mathcal{A} \cap \mathcal{T})\}) \leq C R_X(\mathcal{T})$$

for every  $R$ -bounded set  $\mathcal{T} \subseteq \mathcal{L}(X)$ .

*Proof.* We follow Rozendaal's proof of Theorem 6.3 [1], changing only the uniform geometric input. Put  $\theta = 1 - 1/q$ . Fix  $n \in \mathbb{N}$ , let  $\text{Rad}^n(X) \subset L^2([0, 1]; X)$ , and let  $\tilde{A}$  be the diagonal operator

$$\tilde{A} \left( \sum_{k=1}^n r_k x_k \right) = \sum_{k=1}^n r_k A x_k.$$

Then  $-i\tilde{A}$  generates the diagonal group on  $\text{Rad}^n(X)$ .

By Lemma 2,  $\text{Rad}^n(X)$  is uniformly isomorphic to a complemented subspace of a 1-convex,  $q$ -concave Banach lattice. Hence the operator-valued fractional-domain estimate preceding Theorem 6.2 in [1], applied to  $\tilde{A}$ , gives a constant independent of  $n$  such that

$$\left\| F(\tilde{A}) \right\|_{\mathcal{L}(D((\lambda+i\tilde{A})^\theta), \text{Rad}^n(X))} \leq C_0 \|F\|_{R\mathcal{H}^\infty(\text{St}_\omega; \tilde{\mathcal{A}})}$$

for every admissible operator-valued  $F$ . The independence of  $n$  follows from Rozendaal's Remark 5.3, because all complementability and  $q$ -concavity constants supplied by Lemma 2 are uniform in  $n$ .

Now take  $f_1, \dots, f_n \in R\mathcal{H}^\infty(\text{St}_\omega; \mathcal{A} \cap \mathcal{T})$  and define

$$F(z) \left( \sum_{k=1}^n r_k x_k \right) = \sum_{k=1}^n r_k f_k(z) x_k.$$

Since  $X$  is a complemented subspace of a finite-cotype Banach lattice,  $X$  has property  $(\alpha)$ , as recalled in [1]. The same property- $(\alpha)$  estimate used in the proof of Rozendaal's Theorem 6.2 shows

$$\|F\|_{R\mathcal{H}^\infty(\text{St}_\omega; \tilde{\mathcal{A}})} \leq C_1 R_X(\mathcal{T}),$$

with  $C_1$  independent of  $n$ ,  $f_k$ , and  $\mathcal{T}$ .

Finally, by the diagonal functional calculus,

$$F(\tilde{A}) \left( \sum_{k=1}^n r_k x_k \right) = \sum_{k=1}^n r_k f_k(A) x_k,$$

and

$$D((\lambda + i\tilde{A})^\theta) = \text{Rad}^n(D((\lambda + iA)^\theta))$$

with the natural graph norms. Therefore

$$\left\| \sum_{k=1}^n r_k f_k(A) x_k \right\|_{L^2([0,1]; X)} \leq C_0 C_1 R_X(\mathcal{T}) \left\| \sum_{k=1}^n r_k x_k \right\|_{L^2([0,1]; D((\lambda+iA)^\theta))}.$$

This is exactly the asserted  $R$ -bound. □

**Corollary 1.** *The nontrivial type assumption in Rozendaal’s Theorem 6.3 is not necessary.*

*Proof.* Take, for example,  $X = L^1[0, 1]$ . This Banach lattice is  $q$ -concave for each  $q \geq 2$ , but it has no nontrivial type. Theorem 1 applies to  $X$ , giving the endpoint  $R$ -bounded calculus with exponent  $1 - 1/q$ . Thus the conclusion of the theorem can hold without nontrivial type.  $\square$

## Verification Notes

The proof uses only the source theorem and standard Banach-lattice facts. The new step is Lemma 1; its constants are independent of  $n$ , which is precisely what the original proof needed from  $K$ -convexity. The argument does not claim that every Banach space with property  $(\alpha)$  and finite cotype admits the endpoint theorem. It answers the source question by proving that nontrivial type is not necessary: finite-cotype lattices such as  $L^1$  already provide no-nontrivial-type cases.

## Novelty Check

The run indexes were searched for 1508.02036, Functional calculus for  $C_0$ -groups, second  $R$ -bounded calculus, nontrivial type,  $K$ -convex, Rademacher projection, and related transference/Fourier multiplier phrases. Web searches on 2026-06-20 for exact phrases from the source question and theorem found the source paper and related Rozendaal–Veraar multiplier papers, but no later answer to this endpoint necessity question.

## References

- [1] J. Rozendaal, *Functional calculus for  $C_0$ -groups using (co)type*, arXiv:1508.02036, 2015.
- [2] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces II: Function Spaces*, Ergebnisse der Mathematik und ihrer Grenzgebiete 97, Springer-Verlag, 1979.